

# BST Physics Curriculum Vol 14 Chapter 7 — AC Graph as Theorem Information Network v0.4 (refilled per Cal #104)

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2026-05-23 Saturday EDT (Cal #104 refill)

## Vol 14 Chapter 7 — AC Graph as Theorem Information Network

### Chapter motivation

Mathematical theorem databases organize proved results into knowledge graphs (PolyMath wiki, LeanCommunity, OEIS for integer sequences). BST builds AC (Algebraic Complexity) graph as theorem-information network with ~2200 nodes + ~9850 edges at Saturday EOD; each node = registered theorem (T1-T2482+); edges = derivation dependencies + cross-domain links. AC graph + Grace catalog (4900+ invariants Saturday EOD) + Keeper K-audit chain (K1-K245+) form integrated knowledge infrastructure.

Per Casey CSE directive: AC graph IS substrate-cartography information network at BST research scale. Information-theoretic interpretation: theorem graph carries Shannon entropy (proof-path entropy); depth-cost (AC depth) is information measure; substrate-cartography research = information extraction from graph structure.

### Section 7.0b — Reader-grade 3-level pedagogy

Level 1: AC graph 2200+ nodes + 9850+ edges = theorem-information network; Grace catalog 4900+ invariants + Keeper K-audit chain K1-K245+ integrated knowledge infrastructure; information-theoretic interpretation: depth-cost = information measure; substrate-cartography research = information extraction.

Level 2 (graduate-physicist/information-theorist): Mathematical theorem databases as knowledge graphs: PolyMath wiki collaboration platform, LeanCommunity Lean 4 mathlib formal proof library, OEIS Online Encyclopedia of Integer Sequences, MathSciNet review database. Standard organization: nodes = mathematical objects/theorems; edges = derivation/citation dependencies; graph

operations = forward search, backward search, cross-reference. BST AC (Algebraic Complexity) graph: registered theorems T1-T2482+ as nodes; derivation dependencies + cross-domain bridges as edges; ~2200 nodes + ~9850 edges at Saturday 2026-05-23 EDT. Implementation: `play/ac_graph_data.json` JSON-format graph data; `play/toy_bst_explorer.py` interactive CLI for theorem queries (explore, derive, domain, connect, verify, random, search, stats commands). Grace catalog (4900+ invariants at Saturday EOD; `data/bst_geometric_invariants.json` + `data/bst_constants.json`) provides per-theorem invariant references; Keeper K-audit chain K1-K245+ provides audit-chain governance (Cal Mode 1 referee discipline + tier-label methodology). Per Casey CSE directive (Wednesday May 20, 2026 EOD): AC graph IS substrate-cartography information network at BST research scale; theorem-graph navigation IS substrate-research-as-substrate-computation. Information-theoretic interpretation: theorem-graph carries Shannon entropy  $H(\text{graph}) = -\sum_{\text{path}} p(\text{path}) \log p(\text{path})$  where  $p(\text{path})$  is proof-path probability + bandwidth = average proof-path depth. AC depth-cost framework (Vol 14 Ch 9 cross-link): depth = number of nested AND/OR operations to express theorem; depth  $\leq 2$  universal claim (T29 closed + depth ceiling theorems). Substrate-cartography research = information extraction from graph structure: locate uninvestigated regions via graph topology; identify high-leverage cross-domain bridges via centrality metrics; quantify research progress via graph-coverage analysis. AC graph + catalog + audit chain are integrated knowledge infrastructure for BST substrate-cartography. Cross-link Vol 15 Ch 2 AC graph methodology (Keeper LEAD chapter) for detailed methodology + Vol 14 Ch 9 Kolmogorov complexity + AC(0) compression for theoretical complexity treatment. Casey CSE directive operational realization: BST research IS substrate computational science engineering project; AC graph + catalog + audit chain are operational infrastructure.

Level 3 (5th-grader accessible): Mathematicians use “knowledge graphs” (like Wikipedia for math theorems) to organize proved results. BST has its own AC graph — at Saturday 2026-05-23 EDT, ~2200 theorem nodes + ~9850 dependency edges. Plus Grace’s catalog (4900+ invariants) + Keeper’s audit chain (K1-K245+). All integrated knowledge infrastructure for BST research. Information-theoretic: the theorem graph carries “information” measured in depth (AC depth = how many steps to derive). Casey says BST research IS substrate-cartography via the AC graph.

## Section 7.1 — Standard Math Knowledge Graphs

PolyMath wiki, LeanCommunity mathlib, OEIS integer sequences, MathSciNet.

Standard: nodes = math objects/theorems; edges = derivation/citation dependencies.

## Section 7.2 — BST AC Graph Implementation

~2200 nodes + ~9850 edges at Saturday 2026-05-23 EDT.

play/ac\_graph\_data.json JSON-format; play/toy\_bst\_explorer.py interactive CLI commands (explore, derive, domain, connect, verify, random, search, stats).

### Section 7.3 — Grace Catalog Integration

Grace catalog 4900+ invariants at Saturday EOD; data/bst\_geometric\_invariants.json + data/bst\_constants.json.

Per-theorem invariant references; cross-CI absorption.

### Section 7.4 — Keeper K-Audit Chain (K1-K245+)

K-audit chain provides audit-chain governance: Cal Mode 1 referee discipline + tier-label methodology + cross-CI consensus.

K-audit pre-stages for new theorems + cross-volume sweep for absorption.

### Section 7.5 — Casey CSE Directive AC Graph as Substrate-Cartography

Per Casey CSE directive (Wednesday May 20 EOD): AC graph IS substrate-cartography information network at BST research scale.

Theorem-graph navigation IS substrate-research-as-substrate-computation.

### Section 7.6 — Information-Theoretic Interpretation

Shannon entropy  $H(\text{graph}) = -\sum_{\text{path}} p(\text{path}) \log p(\text{path})$  on proof paths.

Bandwidth = average proof-path depth. AC depth-cost framework (Vol 14 Ch 9):  $\text{depth} \leq 2$  universal claim (T29 closed + depth ceiling theorems).

### Section 7.7 — Substrate-Cartography Research Operations

- Locate uninvestigated regions via graph topology
- Identify high-leverage cross-domain bridges via centrality
- Quantify research progress via graph-coverage analysis

AC graph + catalog + audit chain integrated knowledge infrastructure.

### Section 7.8 — Honest scope + Connection

- AC graph operational at ~2200 nodes + ~9850 edges [\[?\]](#)
- Grace catalog + Keeper K-audit chain integrated [\[?\]](#)
- Casey CSE directive substrate-cartography information network framing
- Open scope: explicit information-theoretic graph-coverage metrics + automated cross-domain bridge identification (multi-month)

Connection: - Vol 15 Ch 2 AC graph methodology (Keeper LEAD chapter) -  
Vol 14 Ch 9 Kolmogorov + AC(0) compression - play/ac\_graph\_data.json +  
play/toy\_bst\_explorer.py - Casey CSE directive Wednesday

— Lyra, Vol 14 Ch 7 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday  
2026-05-23 EDT